Siddhant Jaiswal

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Education

- **Manipal University Jaipur** 2022 – 2026
  B.Tech in Information Technology ; CGPA: 8.45 ***Jaipur, India***
- **Cathedral Sr. Sec. School** 2021 – 2022
  Class XII – Non Medical ; Percentage: 61% ***Lucknow, India***
- **Cathedral Sr. Sec. School** 2019 – 2020
  Class X – CBSE ; Percentage: 81% ***Lucknow, India***

Experience

- **Atthah InfoMedia Pvt. Ltd.** June 2025 – July 2025
  AI Engineer Intern ***Gurgaon,India***
  - Researched and evaluated automation tools such as **N8N** and **ComfyUI** for streamlining repetitive AI workflows.
  - Analyzed lip-syncing and face-swapping models using platforms like **VO3**, **SadTalker**, and other generative AI tools.
  - Conducted feasibility studies comparing open-source and paid solutions across various AI pipelines (image-to-video, text-to-video, video-to-video).
  - Explored AI model performance for Indian languages and storytelling applications.
  - Produced a full-length **AI-generated Hindi story video**, converting a written narrative into 2D-animation-style visuals using MidJourney for image generation, **VO3** for image-to-video synthesis, and audio-sync tooling for narration — targeting regional-language content.
  - Built **ACE Step**, a prompt-based AI music generation system using deep learning models with style transfer and genre-specific composition, generating 50+ tracks across 10+ genre styles with audio post-processing pipelines.

Skills

- **Languages**: Python, C, C++, HTML, CSS
- **Database**: MySQL
- **Tools & Platforms**: GitHub, VS Code, Windows, N8N, ComfyUI, VO3
- **Expertise**: Machine Learning, Automation, Lip-syncing Models
- **Soft Skills**: Adaptability, Eager to Learn, Hard Working, Active Listening, Eloquent Speaker

Projects

- **Multi-Modal Emotion Recognition** Github Link ***April, 2025***
  - **About**: This project implements a multi-modal emotion recognition system that combines facial expressions and speech signals to improve emotion classification. It uses deep learning models for each modality and fuses their features at the feature level for enhanced performance.
  - **Features**: Combines facial expression and speech emotion data for multimodal emotion recognition. Uses CNN-based models for both visual and audio feature extraction. Applies feature-level fusion with Global Average Pooling and dense classification. Improves emotion classification accuracy over unimodal approaches.
  - **Tech Stack**: Python, TensorFlow/Keras, NumPy, OpenCV, Librosa
- **Kiran Traders — Production E-Commerce Platform** Live Site ***June, 2026***
  - **About**: Designed, built and deployed a live wholesale e-commerce platform for a disposable-packaging distributor, covering the full stack from API design to production server administration.
  - **Features**: Built a REST API backend in FastAPI with Pydantic request validation, paired with a React single-page frontend. Integrated the Google Maps Distance Matrix API to compute distance-based delivery charges at checkout, including GCP project and billing configuration. Implemented pincode-based serviceability allow-lists to restrict orders to supported delivery zones. Deployed and maintained the application on a self-managed Debian server with ongoing production debugging and feature releases.
  - **Tech Stack**: FastAPI, React, Python, MySQL, Pydantic, Google Maps Distance Matrix API, Debian/Linux, REST APIs
- **Low-VRAM Automated AI Video Generation Pipeline** ***January, 2026***
  - **About**: An end-to-end automated pipeline that converts structured user input into complete cinematic videos, fully optimized to run on consumer-grade hardware (8GB VRAM, 32GB RAM) instead of high-end GPUs.
  - **Features**: Automated the entire flow from Google Sheets input to final rendered video using n8n orchestration with scheduled triggers, conditional logic, and error handling. Used Ollama (Qwen2.5-7B, Llama 3) to generate structured JSON storyboards with scene numbers, durations, and image/video prompts. Maintained character consistency across scenes via a base reference image with USO reference conditioning. Built three-stage ComfyUI generation: text-to-image, image-to-image scene adaptation, and image-to-video synthesis with LTX 2.0.
  - **Tech Stack**: n8n, ComfyUI, Ollama (Qwen2.5-7B, Llama 3), Flux, LTX 2.0, LoRA, Google Sheets API, FFmpeg, JSON Schema

Achievements

- Published the research paper **"Multi-Modal Emotion Recognition Using Deep Learning Techniques"** in IEEE. [IEEE Xplore]
- Secured a perfect **10.00 GPA** in Semester VIII (Final Semester) of B.Tech Information Technology at Manipal University Jaipur.
- Filed a patent for the **"Low-VRAM Automated AI Video Generation Pipeline"** — a resource-efficient system for automated cinematic video generation on consumer-grade hardware.